

The allegory axioms, and what defines them in the Frobenius calculus

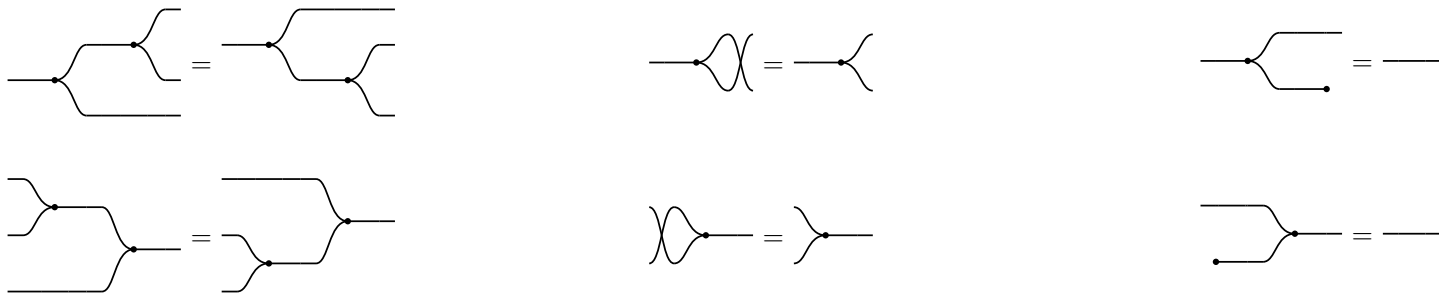
§1. Cartesian bicategory of relations

A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product $\otimes$ (cartisin product of set in Rel, vertical junctposition in the diagram) and

$\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$, **Frobenius**, arrows **lax**.

$(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$ formed a cocommutative comonoid

$(\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ formed a commutative monoid



In $\text{Rel}(\text{Set})$, the four generators are these relations on a set A :

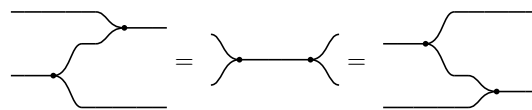
- $\triangleleft = \{(a, (a, a))\}$ — copy: one in, two identical out
- $\triangleright = \{((a, a), a)\}$ — merge: a pair passes only when its two components agree, and the common value comes out
- $\multimap = A \times \{*\}$ — discard: forget the value, keep only that there was one
- $\multimap = \{*\} \times A$ — create: any element at all

§1.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$



The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In Rel it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§1.2. The Frobenius law



The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, $\triangleright \triangleleft$ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, $\triangleleft \triangleright$: copy then merge, a closed loop, which the idempotency of n uses below.

§1.3. Every arrow is a lax comonoid homomorphism

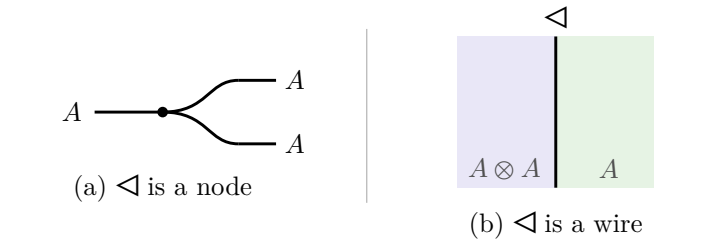


$$R \triangleleft \leq \triangleleft (R \otimes R)$$

$$R \multimap \leq \multimap$$

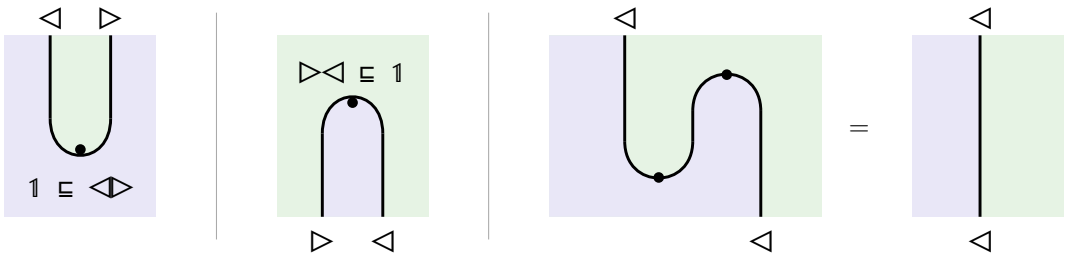
§1.4. The four generators one dimension up

Nothing about $\triangleleft$ changes; what changes is which dimension carries what. Read the same fork in the two calculi — (a) left to right, as every other string diagram here, (b) bottom to top:



	(a) monoidal	(b) 2-cell
region	only one, undrawn	a set: $A, A \otimes A$
wire	a set	a relation: $\triangleleft, \triangleright$
node	a relation: $\triangleleft, \triangleright, \multimap, \multimap$	an inclusion $\sqsubseteq$

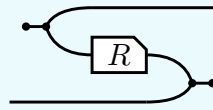
Everything moves up one dimension, so the fork flattens: its three wires were the sets A, A, A , and in (b) sets are regions — the two prongs are no longer drawn at all, they are the name of the region on the left. What (b) gains is the room to draw the inclusions, and with them the adjunction this section opened with:



Green is A , purple $A \otimes A$, and a wire carries its target on the left and its source on the right. The unit lives in A , so A is the inside of the cup; the counit lives in $A \otimes A$. The snake is the adjunction this section opened with, pulled straight.

§2. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

The **converse** R° of $R : a \rightarrow b$ is R with both of its wires turned round,

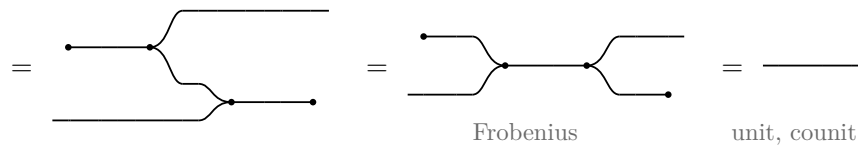


$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. And it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

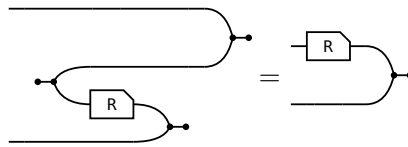
$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§2.1. $1^\circ = 1$ (snake)



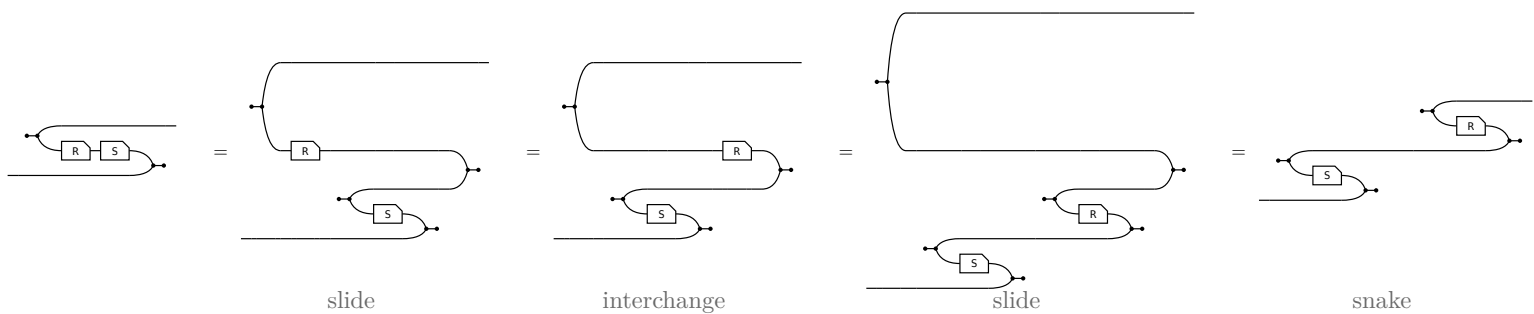
§2.2. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



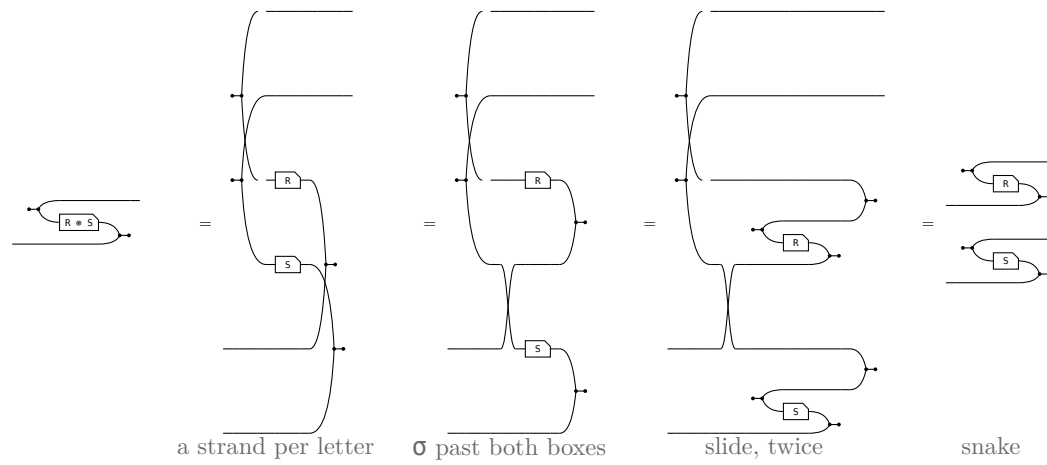
R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circlearrowright$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circlearrowleft \triangleleft$ bends the a strand down and the $\triangleright \circlearrowright$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§2.3. $(R \ S)^\circ = S^\circ R^\circ$



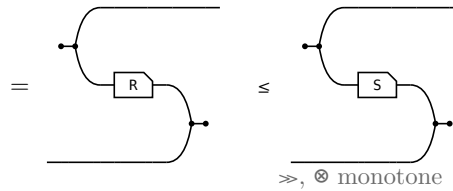
§2.4. $(R \otimes S)^\circ = R^\circ \otimes S^\circ$

A merge at a product is two merges behind a crossing, so the pair unbends one strand at a time and the crossing is all that is left to move.



§2.5. $R \leq S$ implies $R^\circ \leq S^\circ$

R sits in a frame of wires built from $\gg$ and $\otimes$, and both of those are monotone, so the box may be replaced where it stands.



§3. Every arrow is a lax monoid homomorphism

The merge/create mirror of the two comonoid laws opening this note — the paper's (41) and (42), p. 22, which it gets by bending them with (ii) and (iv) above. No bending is needed: expand by the unit $1 \leq \triangleleft \triangleright$, duplicate the box with the comonoid law, contract by the counit $\triangleright \triangleleft \leq 1$. Both halves of the adjunction, and nothing else.

In $\mathbf{Rel}(\mathbf{Set})$ with $R \subseteq A \times B$, where $\triangleright$ is the equality test on a pair and $\bullet$ produces any element at all:

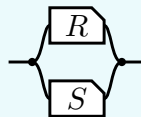
$$\begin{aligned}
 \triangleright R &= \{((a, a), b) : a R b\} \text{ — two equal inputs, then one run of } R \\
 (R \otimes R) \triangleright &= \{((a, a'), b) : a R b \text{ and } a' R b\} \text{ — a run on each input, then the results merged} \\
 \bullet R &= \{(*, b) : \exists a. a R b\} \text{ — the image of } R \\
 \bullet &= \{*\} \times B \text{ — all of } B
 \end{aligned}$$

Take $a' = a$ for the first containment; every b lies in B for the second. Each is strict exactly when its reverse fails — R not injective, R not surjective — which is what the later table on maps reads off them.

inequation, and what proves it	picture
$(41) \triangleright R \leq (R \otimes R) \triangleright$ $\triangleright R = \triangleright R 1 \leq \triangleright R \triangleleft \triangleright \leq \triangleright \triangleleft (R \otimes R) \triangleright \leq (R \otimes R) \triangleright$ — unit, lax $\triangleleft$, counit.	
$(42) \bullet R \leq \bullet$ $\bullet R = \bullet R 1 \leq \bullet R \bullet \bullet \leq \bullet \bullet \bullet \leq \bullet$ — the same three steps at $\mathbb{I}$.	

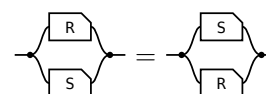
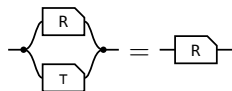
§4. $\cap$ is a commutative idempotent monoid on every hom-set

The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.

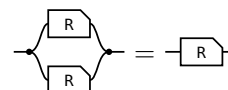
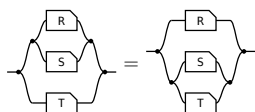


transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).



unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$ **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F (R \cap S) = F R \cap F S$.	

§5. Unitary and pre-tabular — what the other direction needs

condition	allegory	Frobenius side
unitary	τ is a unit when $1 \vdash \tau$ is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes τ exist.	free. $\mathbb{I}$ is the unit object, $1 \leq \multimap \multimap$ says $\multimap$ is entire, and $\tau = \multimap \multimap$ is maximal.
pre-tabular	f, g tabulate R when both are maps, $R = f \circ g$, and R is tabular when some pair tabulates it, the allegory pre-tabular when every arrow lies under a tabular one.	this is what $\otimes$ is. Given unitarity it reduces to tabulating each τ , and a tabulation of τ is an object $n \otimes m$ with its two projections. One side derives the product from tabulations, the other posits it.

Together: **cartesian bicategory of relations** $\simeq$ **unitary pre-tabular allegory**.

§6. Union

Past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure. It draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. The residual, which needs a second composition with linear adjoints, is in the division section below.

statement	picture
$R \sqsubseteq R \cup S$ union $R S := \langle R, S \rangle [1, 1]$: offer both branches, then forget which was taken.	
$R \cup S = S \cup R$	
$\perp \cup R = R$ $\perp$ routes through the zero object; there is no shape for it.	
$R (S \cup T) = R S \cup R T$ — composition distributes over union, which $\cap$ does not.	
$(R \cup S)^\circ = R^\circ \cup S^\circ$	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole distributive , and the one picture with both layers in it.	

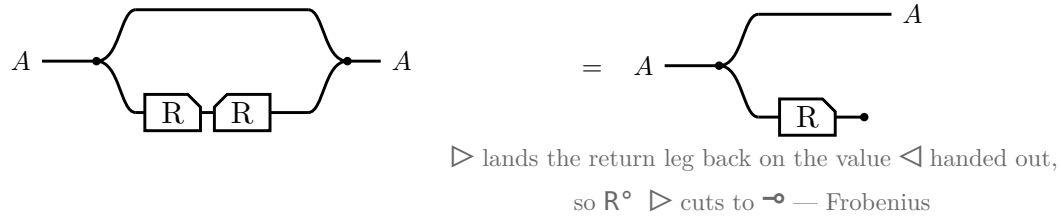
The last of Freyd's §2.21 equations that have no picture here — a tape around nothing, or a tape around what is already inside it, is not worth drawing. R = people who admire a , S = people who admire b throughout.

equation	the reading
$R \cup (S \cap R) = R$ $(R \cup S) \cap R = R$ absorption	People who admire a , or who admire both: still the people who admire a . People who admire a or b , and who admire a : the same. The smaller side of each pair is already inside the larger.

equation	the reading
$R \perp = \perp R$ $0_S = 0_{\{R S\}}$ — and $\perp R = \perp$ on the other side	Admiring someone and then taking a step that leads nowhere gets you nowhere. $\perp$ is a two-sided zero for composition, as τ is not.

§7. Domain and range

The **domain** $\text{Dom } R \triangleq \mathbb{1} \cap R R^\circ$ and the **range** $\text{Ran } R \triangleq \text{Dom } R^\circ$.

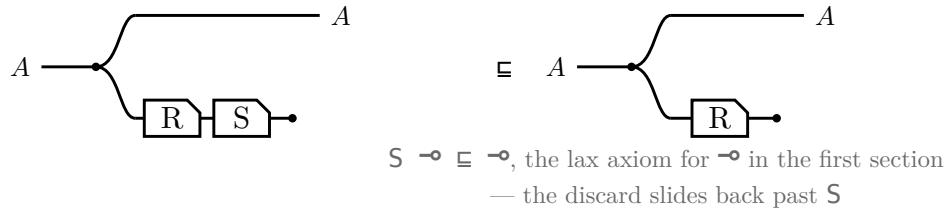


Running R and throwing the result away leaves only the fact that R could fire, and $\text{Ran } R$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

$\text{Dom } R \triangleq \mathbb{1} \cap R R^\circ$
$\text{Dom } R \subseteq A \iff R \subseteq A R, \text{ for } A \text{ coreflexive}$
$\text{Dom } (R S) \subseteq \text{Dom } R$
$\text{Dom } (R \cap S) = \mathbb{1} \cap S R^\circ$
$R \text{ entire} \iff \text{Dom } R = \mathbb{1} \iff \mathbb{1} \subseteq R R^\circ$
$R \text{ simple} \iff R^\circ R \subseteq \mathbb{1}$
$R \text{ a map} \iff R \text{ entire and simple}$
$R, S \text{ entire} \implies R S \text{ entire} \text{ — likewise simple, likewise maps}$
$R S \text{ entire} \implies R \text{ entire}$

§7.1. Sliding the discard

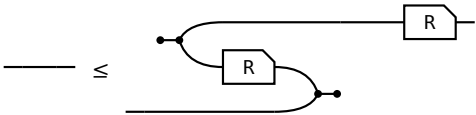
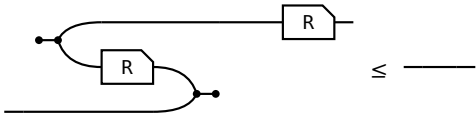
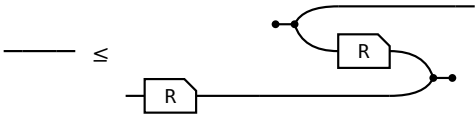
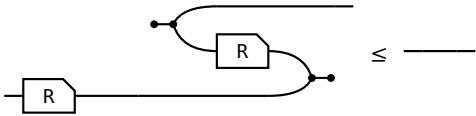
$\text{Dom } (R S) \subseteq \text{Dom } R$, and a single glyph for Dom would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

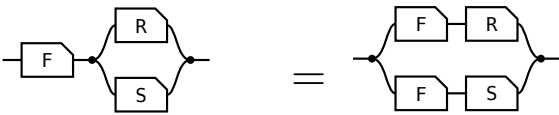


Equality is $S \text{ entire}$, which is the same picture read as $\text{Dom } R = \mathbb{1} \iff R \text{ entire}$.

§8. Maps

Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by the lax monoid section. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory’s **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

 surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	 single valued $R^\circ R \sqsubseteq 1$
 entire $1 \sqsubseteq R R^\circ$. With single valued, a map.	 injective $R R^\circ \sqsubseteq 1$

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = people x may ask, R = admires a, S = admires b. Single valued means one person, so the sides agree.	 one person who admires both a at A, b at B

§9. / is all of

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

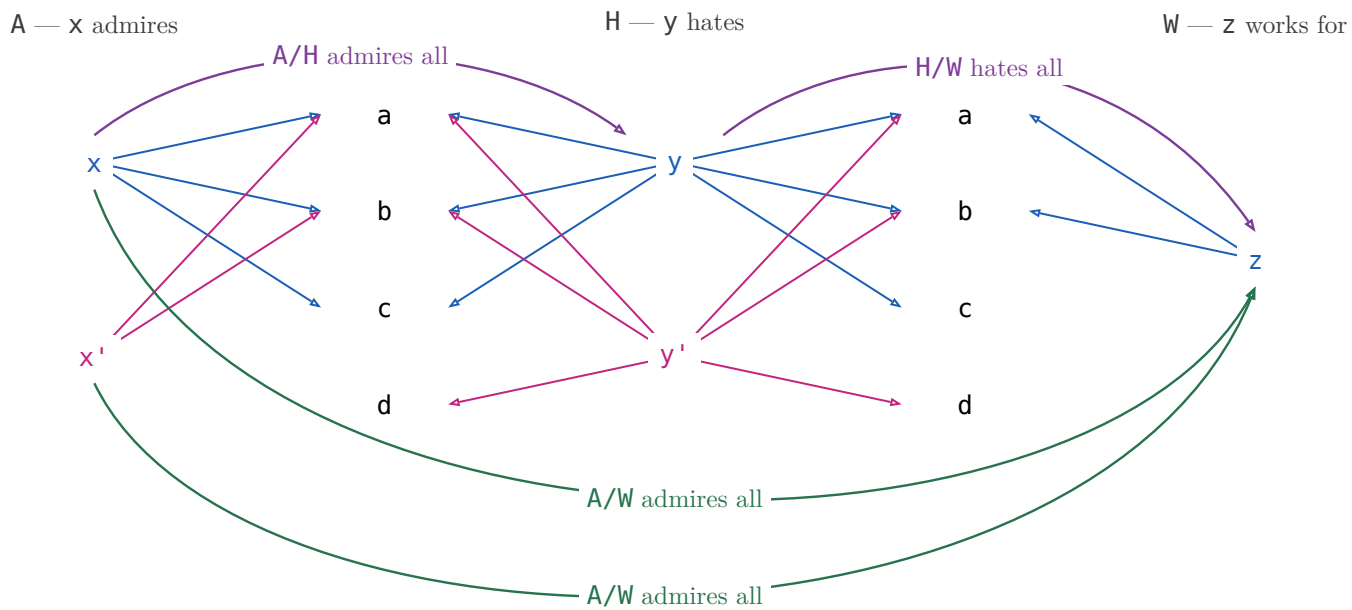
$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/ compares images: $S(y) \subseteq R(x)$. $\setminus$ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§9.1. $(R/S)(S/W) \sqsubseteq R/W$ 

$x (A/H) y$ — x admires everyone y hates

$y (H/W) z$ — y hates everyone z works for

$x (A/W) z$ — x admires everyone z works for

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \sqsubseteq R/S \iff X S \sqsubseteq R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \sqsubseteq S \setminus R \iff S X \sqsubseteq R$ The mirror — divide on the left when x comes first.	
$(R/S) S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S = \emptyset$: R/S is everyone, $(R/S) S = \emptyset$.	
$S (S \setminus R) \sqsubseteq R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	

maps: $f \ (R/S) = (f \ R)/S$ Rename x before or after dividing — the licence to write $f \ R / S$.	
$R/(f \ S) = (R/S) \ f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S) (S/W) \sqsubseteq R/W$ Someone who admires all of a hate-list that already covers everyone z works for admires those people too.	
$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
$(R/R) \ R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's list is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-list is admiring each list in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

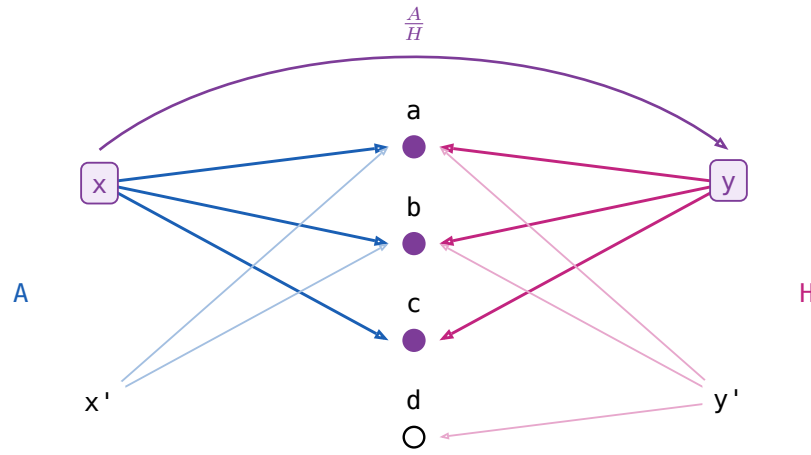
Fifteen laws, fifteen pictures, and not one shows a generator: n , u , $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§10. $x \text{ (Admires\%Hates) } y$ is x admires only all whom y hates

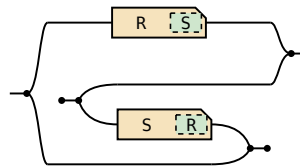
$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

$x \text{ (A/H) } y$ — x admires everyone y hates
 $x \text{ ((H/A)^\circ) } y$ — x admires only people y hates
 $x \text{ ((A/H) \cap (H/A)^\circ) } y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.



A meet of two long divisions, the second turned round by the converse frame:



exported from the definition, not transcribed

$\left(\frac{R}{S}\right)^\circ = \frac{S}{R}$ Matching is symmetric.	
$\frac{R}{S} \frac{S}{W} \sqsubseteq \frac{R}{W}$ And transitive.	

No pictures for the rest of §2.35: symmetric division is not built from the generators.

law	the reading
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R \text{ and } X^\circ R \sqsubseteq S$	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$\frac{R}{S} S \sqsubseteq R$	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S})R$
$\frac{R}{R} R = R$	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y = x$ always qualifies ($1 \sqsubseteq R\%R$ below)
$1 \sqsubseteq \frac{R}{R}$	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$\left(\frac{R}{R}\right)^2 = \frac{R}{R}$	So the relation admires the same people is an equivalence relation.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric X	The largest symmetric arrow that leaves R alone.

law	the reading
$\frac{R}{1}$ is the simple part of R	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{Dom } \frac{R}{S} = 1 \cap (R/S) (S/R)$	Its domain is the domain of simplicity of R .

§10.1. Straight

S is **straight** when $\frac{S}{S} = 1$ — no two y s hate the same people.

law	the reading
every symmetric X with $X \circ S \sqsubseteq S$ is coreflexive	Equivalently, S is straight.
$f \circ S = g \circ S \implies f = g$	A straight S tells its y s apart, so it cancels on the right.
$S \circ R \text{ straight} \implies S \text{ straight}$	If the longer chain tells them apart, the first step already does.
$S \text{ straight} \iff R/S \text{ simple for all } R$	If no two y s agree, at most one x can match a given y .
$R = h \circ S$, h a cover, S straight	In an effective division allegory every arrow factors that way.

§11. Power allegories

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**off, subject to

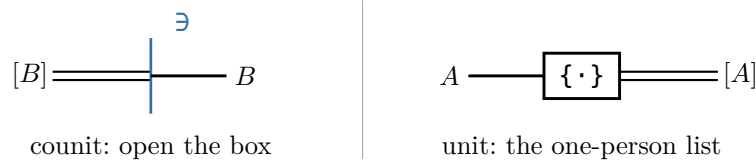
$$\begin{aligned} \exists_R \square &= R \square, \quad \exists_R = \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ (\exists_R/\exists_R) \cap (\exists_R/\exists_R)^\circ &\subseteq \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

$R \square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : [B] \rightarrow B$, dropping the subscript.

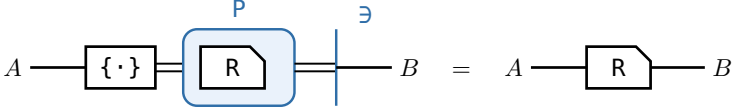
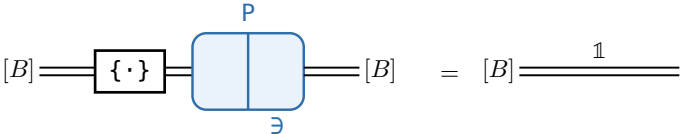
In **Rel** its source is the powerset of its target and $\mathfrak{l} \exists \mathfrak{i}$ iff $\mathfrak{i} \in \mathfrak{l}$, and $\exists$ reads a list $\mathfrak{l}$ back into the people on it.

$\Lambda(R) \triangleq \frac{R}{\exists} : A \rightarrow [B]$, for $R : A \rightarrow B$. The list-of map: send x to the list of the people x admires.

One convention: a $[B]$ -wire is drawn double, because it is a B -wire one level down — inside the box. The box is the power relator P , which the relator section below defines: a box holding $R : B \rightarrow C$ is $P R : [B] \rightarrow [C]$. Inside the box lives the allegory, outside only maps, and $\mathfrak{i} : \mathbf{Map}(\mathcal{A}) \hookrightarrow \mathcal{A}$ is the inclusion that forgets the difference. The adjunction $\mathfrak{i} \dashv P$, that is $\mathcal{A}(A, B) \cong \mathbf{Map}(A, [B])$, has the two ends:



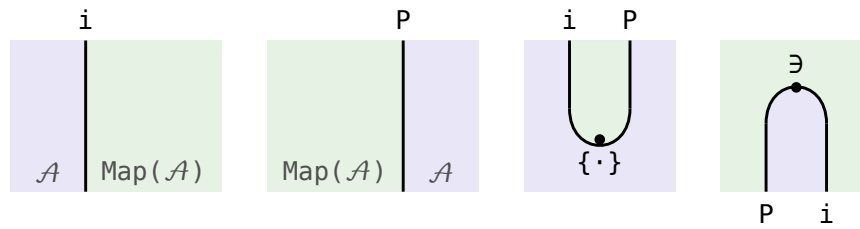
Λ is neither of them — it is the transpose the adjunction gives, $\Lambda(R) = \{\cdot\} (P R)$: make the singleton, then push R through the box.

law	the reading
$\exists_R \square = R \square, \quad \exists_R = \exists_R \square$	$\exists$ has the same target as R , and replacing R by the identity at that target leaves it unchanged: one $\exists$ per object, not per arrow.
$\exists$ is thick	Comprehension : every x has a list of exactly the people x admires. Equivalently every R factors as a map followed by $\exists$.
$\exists$ is straight , that is $\frac{\exists}{\exists} \subseteq \mathbb{1}$	Extensionality : two lists with the same people are the same list.
$\Lambda(R)$ is simple	$\exists$ is straight, and dividing by a straight arrow is simple. At most one list per x .
$\Lambda(R)$ is entire $\iff \exists$ thick	$\text{Dom } \frac{R}{\exists} = \mathbb{1} \cap (R/\exists)(\exists/R)$, the domain row above. At least one list per x , so $\Lambda(R)$ is a map .
$\Lambda(R) \exists = R$ Everything left of the $\exists$ wall is $\Lambda(R)$; open the wall and what was inside comes out unchanged. That is the β of this calculus.	
$\Lambda(R)$ is the only map with $\Lambda(R) \exists = R$	Two maps naming the same people name the same list — extensionality again.
$F \subseteq \Lambda(F \exists)$, F simple	A partial choice of lists is inside the total one.
$[\alpha] = \text{source of } \exists$, the power-object	Every list over α .
$\{\cdot\} \triangleq \Lambda(\mathbb{1})$, the singleton map , monic	The one-person list. $\Lambda(\mathbb{1})\Lambda^\circ(\mathbb{1}) \subseteq \frac{\mathbb{1}}{\exists} \subseteq \frac{\mathbb{1}}{\mathbb{1}} \subseteq \mathbb{1}$.
$\Lambda(\exists) = \mathbb{1}$ Make the list of a list, then read it back one level down. Straightness itself, and one of the two triangle identities of $\mathfrak{i} \dashv P$; the other is $\{\cdot\} \exists = \mathbb{1}$.	

law	the reading
fusion: $\Lambda(f \ R) = f \ \Lambda(R)$, f a map f is a plain rectangle — a map, so it may cross the $\{\cdot\}$ node. That is naturality of the unit, $f \ \{\cdot\} = \{\cdot\} \ (P \ f)$, and it is the whole content of fusion; a chamfered box is stuck on the side it is on.	
$\Lambda(f) = f \ \{\cdot\}$, f a map	Rename first or take singletons first — the fusion row above at $R = 1$.
$\frac{R}{S} = \Lambda(R) \Lambda^\circ(S)$ x and y match exactly when the two boxes hold one and the same list — the $[B]$ -wire joining them is the whole statement.	
$\mathcal{C}$ a topos $\implies \mathbf{Rel}(\mathcal{C})$ a power allegory	And back: a unitary tabular power allegory has $\mathbf{Map}(\mathcal{A})$ a topos. Extensionality and comprehension are all a topos adds.

§12. The same adjunction in Marsden's 2-cell calculus

Other language, other level. Here a **wire** is a functor, a **dot** is a natural transformation, a **region** is a category, and the picture is read bottom to top. Colour marks the category, and for $F : \mathcal{C} \rightarrow \mathcal{D}$ the wire carries the target $\mathcal{D}$ on its left and the source $\mathcal{C}$ on its right (Marsden Def 3.1). R itself cannot appear — it is an arrow, not a 2-cell — which is exactly the trade: this language draws the adjunction, the section above draws the arrows inside it.



the two functors, then the unit $\{\cdot\}$ as a cup and the counit $\exists$ as a cap



left: $\{\cdot\} \ \exists = 1$ on i — the triangle identity the section above only names.
 right: $\Lambda(\exists) = 1$ on P — the straightness of the section above. Both are one wire pulled straight,
 the same shape as the snake in the cartesian-bicategory section with i, P for $\triangleleft, \triangle$.

These two snakes are this note's own snake one bicategory down. $1^\circ = 1$ and **The slide**, in the converse section near the top, pull the same wire straight — there the two bends are $\circ\triangleleft$ and $\triangle\circ$ and the wire is an object of $\mathcal{A}$, here they are $\{\cdot\}$ and $\exists$ and the wire is a functor. The one real difference is how often the situation arises: in $\mathbf{Rel}$ every object is self-dual, so every arrow has its $^\circ$ and the snake costs nothing, while in $\mathbf{Cat}$ an adjoint is rare — $i \dashv P$ is something a power allegory asserts, not something every category has.

What this buys and what it does not. The wall is genuine graphical structure — sealing, opening, and “only maps cross” are the three rules, and $\Lambda(R) \ \exists = R$, $\Lambda(\exists) = 1$, $\Lambda(f \ R) = f \ \Lambda(R)$ are those three rules and nothing more. What it does not buy: the wall is **not** built from $\triangleleft, \circ, \triangle, \circ$. Those four generate $\cap, \cup, ^\circ$ and composition, and the power object is extra structure, posited the same way $/$ is — which is why the fifteen $/$ -pictures in the division section above show black boxes too.

§13. Relator

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S is $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F \mathbb{1} = \mathbb{1} \quad F(R \circ S) = (F R) (F S) \quad R \sqsubseteq S \implies F R \sqsubseteq F S$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

- For f a map, $F f$ is a map and $F(f^\circ) = (F f)^\circ$.
- Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$.
- A relator is fixed by what it does to maps.
- $F(R \cap S) \sqsubseteq (F R) \cap (F S)$, and strictly.
- $F(X \cap Y) = (F X) \cap (F Y)$ for X, Y coreflexive.
- $F(\text{Dom } R) = \text{Dom}(F R)$ for F preserving $\circ$.

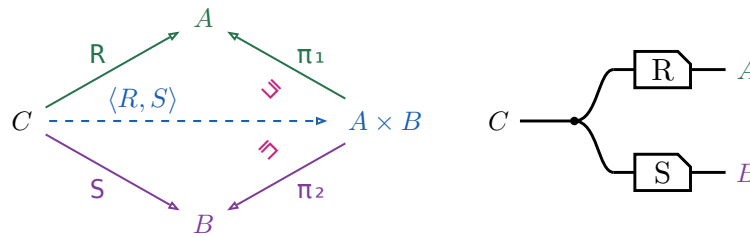
The **power relator** $P \multimap x (P R) y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P R \cap P S$, while $R \cap S = \emptyset$.

§13.1. The pairing $\langle R, S \rangle$

The **pairing** of $R : C \rightarrow A$ and $S : C \rightarrow B$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

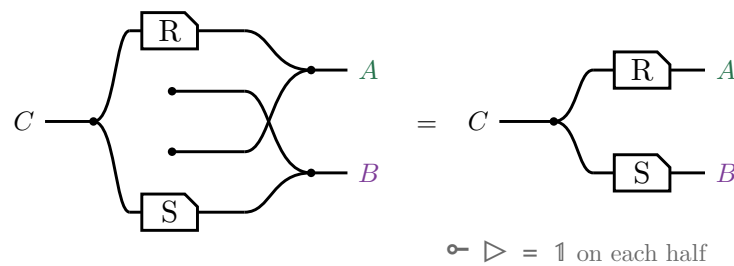
$$\langle R, S \rangle \pi_1 = (\text{Dom } S) R$$

$$\langle R, S \rangle \pi_2 = (\text{Dom } R) S$$



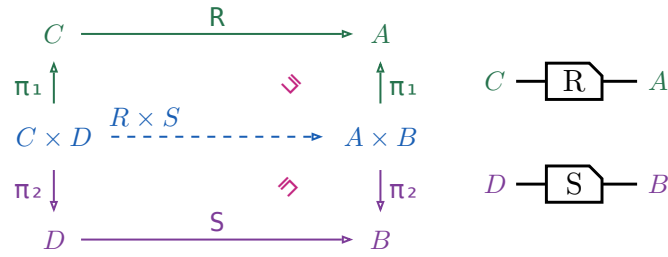
A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In Rel , $c \langle R, S \rangle (a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

No $\circ$ survives the translation. $\pi_1 = \mathbb{1} \otimes \multimap$ discards the second component, so $\pi_1^\circ = \mathbb{1} \otimes \multimap^\circ$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



§13.2. The relational product $R \times S$

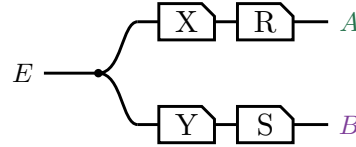
$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.



Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§13.3. Absorption

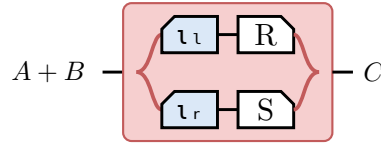
For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle X R, Y S \rangle$. Both sides are this picture:



§13.4. The coproduct $[R, S]$

The injections $\iota_l : A \rightarrow A + B$ and $\iota_r : B \rightarrow A + B$ are maps, and the coproduct they make of the maps stays a coproduct once every arrow is allowed: both equations hold with equality and $[R, S]$ is the only arrow satisfying them, with none of the Dom slack $\langle R, S \rangle$ carries.

$$[R, S] \triangleq \iota_l^\circ R \cup \iota_r^\circ S, \text{ and } R + S \triangleq [R \iota_l, S \iota_r].$$



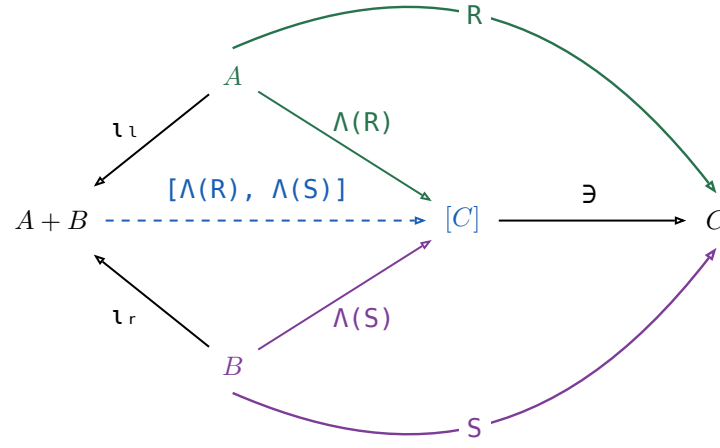
The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.

$\iota_l [R, S] = R, \iota_r [R, S] = S$, and $[R, S]$ is the only such arrow
$\iota_l \iota_l^\circ = \mathbb{1} = \iota_r \iota_r^\circ$
$\iota_l \iota_r^\circ = \perp = \iota_r \iota_l^\circ$
$\iota_l^\circ \iota_l \cup \iota_r^\circ \iota_r = \mathbb{1}$
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$

The first row is not free: ι_l, ι_r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because Λ sends an arrow $A \rightarrow C$ to a map $A \rightarrow [C]$ reversibly, so the map coproduct can be applied underneath it. For any $T : A + B \rightarrow C$,

$$\begin{aligned}
 & \Leftrightarrow \iota_l T = R \text{ and } \iota_r T = S \\
 & \Leftrightarrow \Lambda \text{ is injective — } \Lambda(X) \ni = X \text{ gives } X \text{ back} \\
 & \Leftrightarrow \Lambda(\iota_l T) = \Lambda(R) \text{ and } \Lambda(\iota_r T) = \Lambda(S) \\
 & \Leftrightarrow \Lambda \text{ fusion, } \Lambda(f X) = f \Lambda(X) \text{ for } f \text{ a map} \\
 & \Leftrightarrow \iota_l \Lambda(T) = \Lambda(R) \text{ and } \iota_r \Lambda(T) = \Lambda(S) \\
 & \Leftrightarrow \text{the coproduct of maps, at the map } \Lambda(T) \\
 & \Leftrightarrow \Lambda(T) = [\Lambda(R), \Lambda(S)] \\
 & \Leftrightarrow \Lambda(X) \text{ is the only map with } \Lambda(X) \ni = X \\
 & \Leftrightarrow T = [\Lambda(R), \Lambda(S)] \ni
 \end{aligned}$$

Every step is an $\iff$, so $[\Lambda(R), \Lambda(S)] \ni$ is the **only** arrow satisfying the two equations — which is what $[R, S]$ was claimed to be. Drawn:



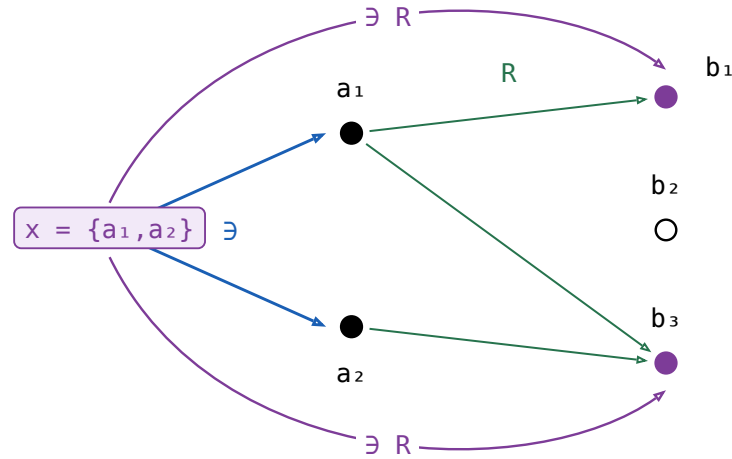
Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the pairing above. The border spells $[R, S] = [\Lambda(R), \Lambda(S)] \ni$, and pushing $\ni$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\iota_l \circ \Lambda(R) \cup \iota_r \circ \Lambda(S)) \ni = \iota_l \circ R \cup \iota_r \circ S$.

§13.5. The power relator $\mathbf{P} R$

For $R : A \rightarrow B$, $\mathbf{P} R \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : [A] \rightarrow [B]$.

$\exists R$ is a **composition** — $\exists : [A] \rightarrow A$ followed by R — and not Freyd's subscripted $\exists_R$, which is the $\exists$ at R 's **target**: §11 writes one $\exists$ per object and drops the subscript. Piece by piece, on one instance, with x , y lists and a , b their elements.

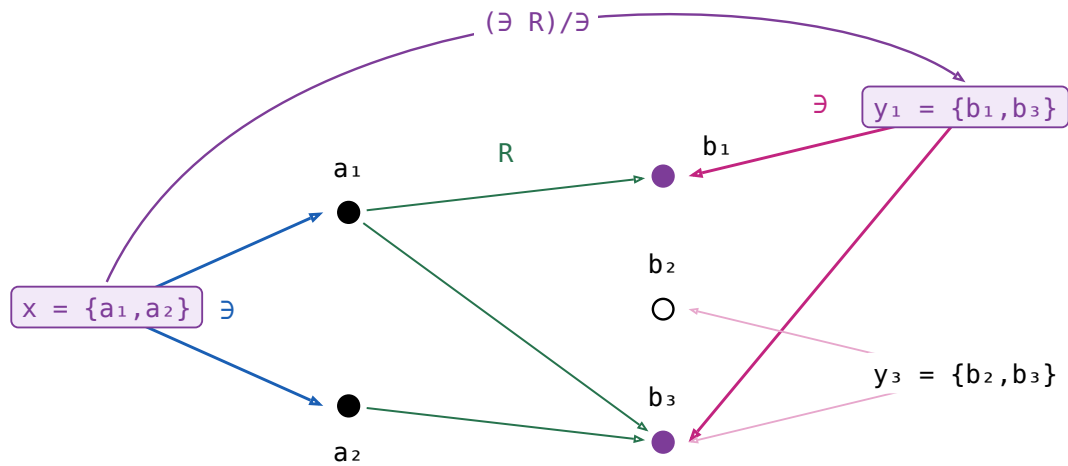
$x = \{a_1, a_2\}$, and R sends a_1 to b_1 and b_3 , a_2 to b_3 alone. Nothing on x reaches b_2 , so b_2 is drawn hollow.



$$\exists R : [A] \rightarrow B \quad x (\exists R) b \iff \exists a \in x. a R b$$

one arc per element of B that x reaches, and b_2 gets none

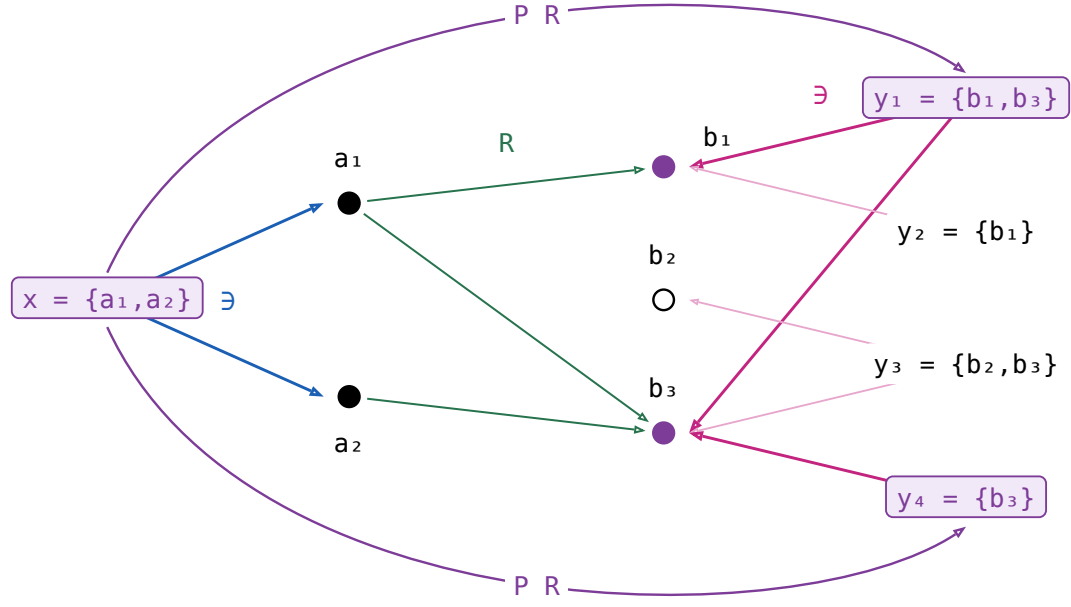
Dividing by $\exists$ turns that into a relation between **lists**: every element of y must be one of the filled dots.



$$(\exists R)/\exists : [A] \rightarrow [B] \quad x ((\exists R)/\exists) y \iff \forall b \in y. \exists a \in x. a R b$$

y_3 is rejected: it names b_2 , which x does not reach

The other half of the meet is that clause with the lists swapped and R turned round: every element of x must reach y . Both together, and two more lists to separate them:



$$((\exists R^\circ)/\exists)^\circ : [A] \rightarrow [B] \quad x ((\exists R^\circ)/\exists)^\circ y \iff \forall a \in x. \exists b \in y. a R b$$

y_2 is rejected too: a_2 's only image is b_3 , which y_2 does not name

In words, if $x (P R) y$ then every element of x is related by R to some element of y , and conversely — the two clauses of n , one per direction.

Neither $\exists$ nor $P R$ is a map. $\exists$ sends a list to **each** of its elements; $P R$ sends x to **every** y meeting the two clauses, y_1 and y_4 both. Nor is it entire: an $a \in x$ with no image at all leaves x with no partner whatever. Turning a relation into a map is Λ 's job (§11), and on lists that map is $\Lambda(\exists R)$ — of the four it accepts y_1 only.

Not a symmetric division. $\frac{\exists R}{\exists}$ is $\Lambda(\exists R)$, and in $\mathbf{Rel}$ it reads $x \Lambda(\exists R) y \iff y = \{b \mid \exists a \in x. a R b\}$: in words, y is **exactly** what x reaches, where $P R$ asks only that each side cover the other. Nor can it be repaired: in that fraction's second half $(\exists / (\exists R))^\circ$ the R sits in a denominator, so the operation is antitone there, and a relator has to be monotone. $P R$ keeps the first half and takes the second half at R° , which puts R back in a numerator. The fraction returns exactly where the two halves agree — at 1 , and at a map.

law	the reading
$X \sqsubseteq P R \iff X \exists \sqsubseteq \exists R$ and $X^\circ \exists \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = (P R)^\circ$, and $R \sqsubseteq S \implies P R \sqsubseteq P S$.
$P 1 = \frac{\exists}{\exists} = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$P f = \Lambda(\exists f) = \frac{\exists f}{\exists}$, for f a map	In $\mathbf{Rel}$, $x (P f) y \iff y = \{f a \mid a \in x\}$. The half at f° says every $a \in x$ has its $f a$ on y ; f has just the one image per a , so that already says y contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R S) = (P R)(P S)$	$\sqsubseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R S)$.

§14. Catamorphism

F a relator with an **initial algebra** $\alpha : F T \rightarrow T$ among the maps. For a relational algebra $R : F B \rightarrow B$, the **catamorphism** $\langle R \rangle : T \rightarrow B$ is the unique arrow with $\alpha \langle R \rangle = (F \langle R \rangle) R$.

$$\begin{array}{ccc}
 F T & \xrightarrow{F X} & F B \\
 \alpha \downarrow & & \downarrow R \\
 T & \xrightarrow{X} & B
 \end{array}$$

With $X = \langle R \rangle$ the square commutes strictly: unlike the product's triangles, there is no slack here, so no $\sqsubseteq$ appears anywhere in it.

The three **fusion** rows rewrite $\langle R \rangle S$ through a second algebra $Q : F C \rightarrow C$ along an arrow $S : B \rightarrow C$. They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest of the table needs only the initial algebra.

name	law	what it says
the defining equation	$X = \langle R \rangle \iff \alpha X = (F X) R$	Folding a value that α has just built is the same as folding its parts and combining them with R . Nothing else has that property.
Lambek	$\alpha^\circ = \alpha^{-1}$, the initial algebra is an isomorphism	A constructor can be undone — α° takes a value apart into the parts it was built from.
Eilenberg–Wright	$\Lambda \langle R \rangle = \langle \Lambda((F \exists) R) \rangle$	A relational fold is a deterministic fold of SETS: Λ pushes the nondeterminism into the power-object, where the fold is a map again.
Eilenberg–Wright	$\langle R \rangle = \langle \Lambda((F \exists) R) \rangle \exists$	The same fact read back — fold deterministically into a set, then take a member of it.
fusion	$\langle Q \rangle \sqsubseteq \langle R \rangle S \iff (F S) Q \sqsubseteq R S$	Half of fusion: an inclusion between the two algebras is inherited by the folds.
fusion	$\langle R \rangle S \sqsubseteq \langle Q \rangle \iff R S \sqsubseteq (F S) Q$	The other half, with both inclusions turned around.
fusion	$\langle R \rangle S = \langle Q \rangle \iff R S = (F S) Q$	Both halves at once: a fold followed by S collapses into a single fold, which is how an intermediate structure is got rid of.

The two Λ rows, drawn:

$$\begin{array}{ccccc}
 F T & \xrightarrow{F K} & F [B] & \xrightarrow{F \exists} & F B \\
 \alpha \downarrow & & \downarrow \Lambda((F \exists) R) & & \downarrow R \\
 T & \xrightarrow{K} & [B] & \xrightarrow{\exists} & B
 \end{array}$$

The left square is that catamorphism's own defining square, $K = \langle \Lambda((F \exists) R) \rangle$, and the right one is Λ 's cancellation, so the outer rectangle says $K \exists$ satisfies the defining equation of $\langle R \rangle$ — and uniqueness finishes it.

§15. Type functor

F a **binary** relator: $F(R, S)$ is its action on a pair, and $F X$ abbreviates $F(1, X)$, the F of the catamorphism section. For every object A the initial algebra is $\alpha : F(A, T A) \rightarrow T A$, among the maps. The **type functor** T acts on an arrow $R : A \rightarrow B$ by

$$T R = (F(R, 1) \alpha) : T A \rightarrow T B$$

F is the **base functor** — one layer of the structure, acting on the recursive position — while T is the datatype itself, acting on the parameter. For cons-lists, `list R = ([nil, (R @ 1) cons])`, which is `map R`.

$$\begin{array}{ccccc}
 & & F(1, T R) & & F(R, 1) \\
 F(A, T A) & \xrightarrow{\quad\quad\quad} & F(A, T B) & \xrightarrow{\quad\quad\quad} & F(B, T B) \\
 \downarrow \alpha & & & & \downarrow \alpha \\
 T A & \xrightarrow{\quad\quad\quad T R \quad\quad\quad} & T B & &
 \end{array}$$

$T R$ is the unique arrow making it commute; there is no $\sqsubseteq$ in it.

name	law	what it says
the defining equation	$T R = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way; that is <code>map R</code> .
functor	$T 1 = 1$ and $(T R)(T S) = T (R S)$	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	$(T R) (Q) = (F(R, 1) Q)$	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of α	$\alpha (T R) = F(R, T R) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G R = F(R, T R)$ to T .
type relator	$(T R)^\circ = T (R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion row above applied to $T R$'s own defining algebra, whose side condition holds because F is a bifunctor — $F(R, 1) F(1, (Q)) = F(R, (Q)) = F(1, (Q)) F(R, 1)$ — so it inherits that row's local-completeness requirement.

$$\begin{array}{ccc}
 F(A, T A) & \xrightarrow{\quad F(R, T R) \quad} & F(B, T B) \\
 \downarrow \alpha & & \downarrow \alpha \\
 T A & \xrightarrow{\quad T R \quad} & T B
 \end{array}$$

It commutes strictly: it is the naturality square of α .